Introduction to Artificial intelligence

Tutorial Sheet 7 (Logic and Inference - Solution)

Exercise 1

A. Rewrite the following statements in First-Order Logic. You may use appropriate predicates. The used predicates should reasonably and logically make sense. For each sentence, I have obviously considered the ones below, or logically equivalent ones.

i. Every coyote chases some roadrunner.
$\forall x (COYOTE(x) \Rightarrow \exists y (RR(y) \wedge CHASE(x,y)))$ or
$\forall x \exists y (COYOTE(x) \wedge RR(y) \wedge CHASE(x,y))$

ii. Every roadrunner who says ``beep-beep" is smart.
Here, one can use BEEP(x) or Say(x, BeepBeep), etc.
$\forall x (RR(x) \wedge BEEP(x) \Rightarrow SMART(x))$

iii. No coyote catches any smart roadrunner.
I have accepted two possible readings of the sentence. The following is one of them. Obviously, I took this into account in the conversion to CNF.
$\neg \exists x \exists y (COYOTE(x) \wedge RR(y) \wedge SMART(y) \wedge CATCH(x,y))$

iv. Any coyote who chases some roadrunner but does not catch it is frustrated.
$\forall x (COYOTE(x) \wedge \exists y (RR(y) \wedge CHASE(x,y) \wedge \neg CATCH(x,y)) \Rightarrow FRUSTRATED(x))$

B. Convert the statements in CNF showing all the steps.

i. $\forall x (COYOTE(x) \Rightarrow \exists y (RR(y) \wedge CHASE(x,y)))$
- Eliminate Implication:
  $\forall x (\neg COYOTE(x) \vee \exists y (RR(y) \wedge CHASE(x,y)))$
- Skolemise: substitute y by Skolem function K(x)
  $\forall x (\neg COYOTE(x) \vee [RR(K(x)) \wedge CHASE(x, K(x))])$
- Distribute $\vee$ over $\wedge$:
  $\forall x (\neg COYOTE(x) \vee RR(K(x))) \wedge (\neg COYOTE(x) \vee CHASE(x, K(x)))$
- Drop the universal quantifiers $\forall$
  $(\neg COYOTE(x) \vee RR(K(x))) \wedge (\neg COYOTE(x) \vee CHASE(x, K(x)))$
- Eliminate Conjunction :
  $\neg COYOTE(x) \vee RR(K(x))$
  $\neg COYOTE(x) \vee CHASE(x, K(x))$

ii. $\forall x (RR(x) \wedge BEEP(x) \Rightarrow SMART(x))$
- Eliminate Implication:
  $\forall x (\neg RR(x) \vee \neg BEEP(x) \vee SMART(x))$
- Drop the universal quantifier $\forall$
  $\neg RR(x) \vee \neg BEEP(x) \vee SMART(x)$

iii. $\neg \exists x \exists y (COYOTE(x) \wedge RR(y) \wedge SMART(y) \wedge CATCH(x,y))$
- Apply De Morgan's law to push negation inward:
  $\forall x \forall y \ \neg (COYOTE(x) \wedge RR(y) \wedge SMART(y) \wedge CATCH(x,y))$
- Distribute $\neg$ over $\wedge$:
  $\forall x \forall y \ \neg COYOTE(x) \vee \neg RR(y) \vee \neg SMART(y) \vee \neg CATCH(x,y)$
- Drop the universal quantifier $\forall$
  $\neg COYOTE(x) \vee \neg RR(y) \vee \neg SMART(y) \vee \neg CATCH(x,y)$

iv. $\forall x (COYOTE(x) \wedge \exists y (RR(y) \wedge CHASE(x,y) \wedge \neg CATCH(x,y)) \Rightarrow FRUSTRATED(x))$
- Eliminate Implication:
  $\forall x \neg (COYOTE(x) \wedge \exists y (RR(y) \wedge CHASE(x,y) \wedge \neg CATCH(x,y)) \vee FRUSTRATED(x))$

– Distribute ¬ over first ∧ :
∀ x ¬COYOTE(x) ∨ ¬ ∃y (RR(y) ∧ CHASE(x,y) ∧ ¬ CATCH(x,y)) ∨ FRUSTRATED(x))

– Apply De Morgan's law to push negation inward:
∀ x ¬COYOTE(x) ∨ ∀y ¬(RR(y) ∧ CHASE(x,y) ∧ ¬ CATCH(x,y)) ∨ FRUSTRATED(x))

– Distribute ¬ over remaining ∧ :
∀ x ¬COYOTE(x) ∨ ∀y ¬RR(y) ∨ ¬CHASE(x,y) ∨ ¬ ¬ CATCH(x,y)) ∨ FRUSTRATED(x))

– Remove ¬ ¬
∀ x (¬COYOTE(x) ∨ ∀y ¬RR(y) ∨ ¬CHASE(x,y) ∨ CATCH(x,y)) ∨ FRUSTRATED(x)

– Drop the universal quantifiers ∀
¬COYOTE(x) ∨ ¬RR(y) ∨ ¬CHASE(x,y) ∨ CATCH(x,y) ∨ FRUSTRATED(x)

Exercise 2 (Taken from Tutorial 6):

A. Rewrite the following statements in First-Order Logic. Choose appropriate predicates and constants.

1. All dogs howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. Tariq has either a cat or a dog.
5. (Conclusion) If Tariq is a light sleeper, then Tariq does not have any mice.

B. Convert the statements in CNF showing all the steps.

C. Apply the Resolution inference rule to prove the conclusion (statement 5) from the first four statements (statements 1 to 4). Make sure that each step all the substitutions are shown as well as the parent clauses and the resolvent.

Solutions:

A.

1. $\forall x\ (DOG(x) \rightarrow HOWL(x, Night))$
2. $\forall x\ \forall y\ (HAVE(x,y) \wedge CAT(y) \rightarrow \neg\ \exists z\ (HAVE(x,z) \wedge MOUSE(z)))$
3. $\forall x\ (LS(x) \rightarrow \neg\ \exists y\ (HAVE(x,y) \wedge HOWL(y, Night))$
4. $\exists x\ (HAVE(Tariq,x) \wedge (CAT(x) \vee DOG(x)))$
5. $LS(Tariq) \rightarrow \neg\ \exists z\ (HAVE(Tariq,z) \wedge MOUSE(z))$

B.

1. $\forall x\ (DOG(x) \rightarrow HOWL(x, Night))$
   $\neg\ DOG(x) \vee HOWL(x, Night)$

2. $\forall x\ \forall y\ (HAVE(x,y) \wedge CAT(y) \rightarrow \neg\ \exists z\ (HAVE(x,z) \wedge MOUSE(z)))$
   $\forall x\ \forall y\ (HAVE(x,y) \wedge CAT(y) \rightarrow \forall z\ \neg\ (HAVE(x,z) \wedge MOUSE(z)))$

   $\forall x\ \forall y\ \forall z\ (\neg\ (HAVE(x,y) \wedge CAT(y)) \vee \neg\ (HAVE(x,z) \wedge MOUSE(z)))$

   $\neg\ HAVE(x,y) \vee \neg\ CAT(y) \vee \neg\ HAVE(x,z) \vee \neg\ MOUSE(z)$

3. $\forall x\ (LS(x) \rightarrow \neg\ \exists y\ (HAVE(x,y) \wedge HOWL(y, Night))$
   $\forall x\ (LS(x) \rightarrow \forall y\ \neg\ (HAVE(x,y) \wedge HOWL(y, Night))$

   $\forall x\ \forall y\ (LS(x) \rightarrow \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night))$

   $\forall x\ \forall y\ (\neg\ LS(x) \vee \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night))$

   $\neg\ LS(x) \vee \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night)$

4. $\exists x\ (HAVE(Tariq,x) \wedge (CAT(x) \vee DOG(x)))$

HAVE(Tariq,A) ∧ (CAT(A) ∨ DOG(A))

5. ¬ *[LS(Tariq) → ¬ ∃z (HAVE(Tariq,z) ∧ MOUSE(z))] (negated conclusion)*
   ¬ *[¬ LS (Tariq) ∨ ¬ ∃z (HAVE (Tariq, z) ∧ MOUSE(z))]*

   LS(Tariq) ∧ ∃z (HAVE(Tariq, z) ∧ MOUSE(z)))

   LS(Tariq) ∧ HAVE(Tariq,B) ∧ MOUSE(B)

   Or you may give the converted sentence here and negate it below, yielding the 3 clauses under 5 below.

The set of CNF clauses for this problem is thus as follows:

1. ¬ *DOG(x) ∨ HOWL(x, Night)*
2. ¬ *HAVE(x,y) ∨ ¬ CAT(y) ∨ ¬ HAVE(x,z) ∨ ¬ MOUSE(z)*
3. ¬ *LS(x) ∨ ¬ HAVE(x,y) ∨ ¬ HOWL(y, Night)*
4. a. *HAVE(Tariq,A)*
   b. *CAT(A) ∨ DOG(A)*
5. a. *LS(Tariq)*
   b. *HAVE(Tariq,B)*
   c. *MOUSE(B)*

C.
The resolution-based proof tree is being presented under a tabular format (just for convenience of not having to draw a tree):

Parent clause numbers	**Resolvent Number**	***Resolvent***	***Applied Substitution***
[1 4.b]	6.	*CAT(A) ∨ HOWL(A, Night)*	*{x/A}*
[2 5.c]	7.	¬ *HAVE(x,y) ∨ ¬ CAT(y) ∨ ¬ HAVE(x,B)*	*{z/B}*
[7 5.b]	8.	¬ *HAVE(Tariq,y) ∨ ¬ CAT(y)*	*{x/Tariq}*
[6 8]	9.	¬ *HAVE(Tariq,A) ∨ HOWL(A, Night)*	*{y/A}*
[4.a 9]	10.	*HOWL(A, Night)*	*{}*
[3 10]	11.	¬ *LS(x) ∨ ¬ HAVE(x,A)*	*{y/A}*
[4.a 11]	12.	¬ *LS(Tariq)*	*{x/Tariq}*
[5.a 12]	13.	□	*{}*

Empty clause reached, QED.

Exercise 3:

1. Translate the following sentences into for the First-Order Logic sentences with an appropriate encoding:

a) There are three children Ali, Boudour and Laith.
(Hint: you must explicitly ensure that different names refer to distinct individuals).

b) Every child is wearing either a hat, coat or a scarf.
c) Ali is wearing a red hat.
d) Boudour is wearing a green coat.
e) Laith is wearing a blue scarf.
f) Ali has a favourite colour, which is either red, green, or blue.
g) Laith's favourite colour is green.
h) Boudour's favourite colour is red.

Solution

1) First-order logic encodings for the given statements:
a) $\exists x\, \exists y\, \exists z\, (x \neq y \wedge x \neq z \wedge y \neq z \wedge \text{Child}(x) \wedge \text{Child}(y) \wedge \text{Child}(z) \wedge \text{Name}(x,\text{Ali}) \wedge \text{Name}(y,\text{Boudour}) \wedge \text{Name}(z,\text{Laith}))$
b) $\forall x\, (\text{Child}(x) \rightarrow (\text{Wearing}(x,\text{Hat}) \vee \text{Wearing}(x,\text{Coat}) \vee \text{Wearing}(x,\text{Scarf})))$
c) Wearing(Ali,RedHat)
d) Wearing(Boudour,GreenCoat)
e) Wearing(Laith,BlueScarf)
f) ([FavoriteColor(Ali,Red) ∧¬FavoriteColor(Ali,Green) ∧ ¬FavoriteColor(Ali,Blue)] ∨[FavoriteColor(Ali,Green) ∧¬FavoriteColor(Ali,Red) ∧ ¬FavoriteColor(Ali,Blue)]∨[FavoriteColor(Ali,Blue) ∧¬FavoriteColor(Ali,Green) ∧ ¬FavoriteColor(Ali,Red)])
g) FavoriteColor(Laith,Green)
h) FavoriteColor(Boudour,Red)
2) To infer whether "No child (Laith and Boudour)is wearing anything that has his/her favorite color" using forward chaining, we need to derive this statement from the given premises. We can attempt to do this by considering each child and his/her favorite color:

